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Re-thinking the cultural codes of new media: The question concerning ontology

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Abstract

The digital world need not solely be conceived in Western, elite terms, but instead can and should be re-envisioned as a space that empowers the values, priorities, and ontologies held by global users from the 'margins', within the developing world. This paper asks us to re-consider the ontologies of the digital world in light of the cultural beliefs, languages, and value systems of emerging web users. I argue that as we design new media technologies and develop projects we must think about local ontologies and practices rather than singular Western-created representations of knowledge. I present several examples that think past rigid, hierarchical classifications, opening up the codes of new media to better listen to diverse community and cultural voices.

Keywords

Algorithms, digital culture, diversity, fluid ontology, global, indigenous knowledge, new media, ontology, power, web 2.0

Introduction

As those connected by the increasingly global Internet exceeds 2 billion, users from the 'margins', economically and within the developing world, hold the potential to dramatically influence research on digital cultures, particularly around the question of whose voices drive the architectures, algorithms, and languages of new media. Yet most existing web technologies are produced, designed, and built for Western (and increasingly East Asian) audiences, implying that users from the Global South may remain a silent majority. Algorithms, such as Google's PageRank, reaffirm

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these realities by emphasizing sites that are more widely known and thus supposedly socially credible (Introna and Nissenbaum, 2000; Vaidyanathan, 2011). They operate on the understanding that credibility can be calculated around mass validated links, yet fail to point out that validation is a socially constructed phenomenon, biased by those that control infrastructure, wealth, literacy, and networks. As Google and other search engines export their algorithms, these ways of knowing and ordering the world may not only impose themselves onto peoples within the developing world, but also present a stratified view of knowledge favoring those with power and wealth. Asking technologists to 'give us some control' in a recent TED talk (2011a), Pariser notes in a recent monograph (2011b: 9–19) that the driving force behind social media technologies such as Yahoo News, Facebook, and Google is user-based personalization.

Like algorithms, hierarchical databases are the key building block of many web-based systems. Scholars argue these technical instruments are the evolution of an Enlightenment-era ideal that exerts power and control through the logic of positivism and calculability (Adorno and Horkheimer, 1947; Bowker, 2005). Hierarchies and standards emerge out of a long-standing history of scientific abstraction, a separation between knower and known which shapes information according to stabilized terms and ontologies that obviate other diverse cultural ways of knowing, speaking, and narrating. Turnbull (n.d) has explained that trying to preserve such knowledge according to a single common standard that emerges is a trap of Western thinking that ironically obliterates the diversity of the traditions from which these practices of knowledge emerge. While technologies built with elite and Western social sensibilities are increasingly appropriated and actively used in diverse ways by users and cultures worldwide, they remain engineered with what Feenberg (2005: 52) calls a 'technical code', or the implementation of a particular interest, value, or ideology in the engineering of a technology.

Though widely considered a difficult term to define, as explicated by Sen's (2006: 113) discussion of the differences between cultural conservation versus cultural liberty, many agree that its core 'culture' refers to diverse ways of knowing and being, which this article refers to as ontologies (Srinivasan, 2006b). Ontologies refer to the modes by which knowledge is articulated, expressed, interpreted, and formalized. An ontological turn in new media studies reminds us that nature, science, and technology are constructed in relation to the web of networks, actors, and relationships that define them (Law and Mol, 2008; Woolgar and Pawluch, 2004). These networks of construction naturally resonate with cultural and community diversity, as different communities communicate, interpret, and socially construct ontologies based on their diverse social, cultural, and political goals. Considering diverse communities as active authors, hackers, and designers of new media present possibilities to design and deploy new media technologies based on multiple, diverse ontologies.

In contrast, most ontologies populating the digital world emerge from the monocultures of Western corporations and cultural institutions, such as museums and libraries (Boast et al., 2007). The ontologies that anchor these systems emerge out of practices that Suchman (2003: 5) terms 'detached intimacy', whereby technologies are detached from their increasingly global audiences and tethered to organizational and cultural histories and standards. User-statistics research affirms this monocultural approach toward ontology, by solely studying use in prefigured, quantifiable manners, without considering alternate ontological approaches toward evaluation, appropriation, or design.

This article steps away from these homogenizing shackles to re-consider the ontologies of the digital world in light of the cultural beliefs, languages, and value systems of emerging web users. Inspired by Deleuze and Guattari's (1987: 21) discussion of the rhizome as an alternative to the hegemonies of fixed, hierarchical, dominating modes of constructing and articulating knowledge, I believe we must also embrace multiplicity in our thinking around global new media studies. This can be inspired by thinking about knowledge as embodied, performed, and narrated, experiential rather than symbolic, and resonant with ontological traditions practiced by non-Western local, rural, and indigenous peoples, though of course neither exclusively or completely (Turnbull, n.d; Verran, 2002). Opening up the conversation about ontology in the digital world foregrounds ethical questions about the sovereignty of diverse knowledge, and whether the voices of emerging users should be ignored or empowered. Without this turn toward ontological multiplicity, new media technologies maintain rather than interrupt historical patterns (Gitelman, 2006) that fail to interrogate the 'inscription technologies' (Hayles, 2002: 25) which indirectly produce hegemonic modes of thinking and ordering.

By moving away from a limited discussion around information access toward considering the power of appropriation and grassroots authorship, this article argues that in partnership with diverse communities scholars can interrogate and subvert the logic and underlying ontologies of systems themselves, building on a rich history that explores the social construction of technology and systems (Bijker and Law, 1992; Pinch, 1992; Suchman 2003). This article thus presents several approaches toward creating and designing systems that consider culturally diverse ontologies. This trajectory is described in Figure 1 below.

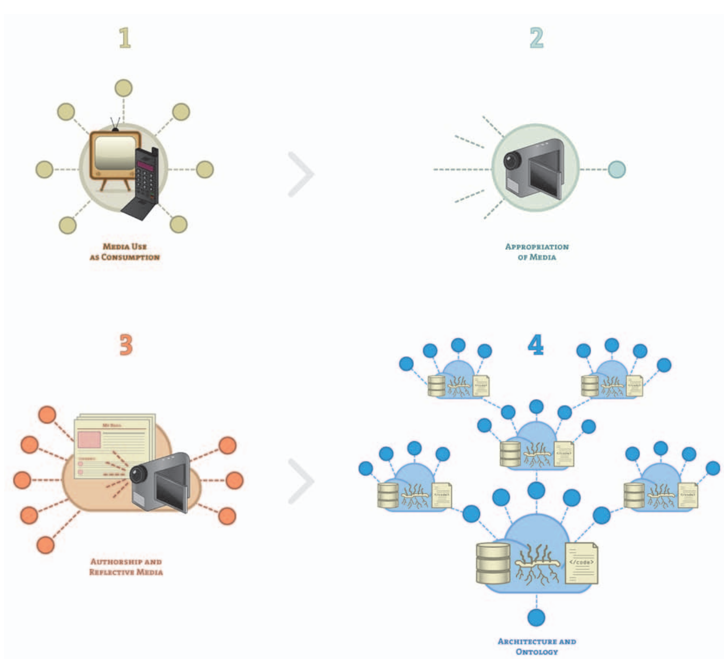

Figure 1. A focus on cultural voices shifts new media studies as above.

A flattening of the world – considering techne and access

Digital technologies emerge from a historical desire to make processes more efficient and routinized (Taylor, 1911), extracting signals from noise (Shannon and Weaver, 1949). Yet instead of primarily discussing technologies teleologically as tools to accomplish tasks, Heidegger (1977) reminds us in *The Question Concerning Technology* that technologies also reveal underlying ontological beliefs around how the world should be ordered. These ontologies can speak to different 'laws of cool' that avoid the 'creative destruction' of postindustrial approaches and instead focus on historical and cultural knowledge (Liu, 2004). For Heidegger, the Greek term techne, the root of technology, refers not only to products and tools but also the poetics and aesthetics of the mind. Techne reveals the ways in which human societies make and articulate knowledge, speaking to social poetics and cognition.

Yet as illustrated by Adorno and Horkheimer (1947), technologies are but a part of a history whereby Enlightenment ideals have self-destructed to empower the apparatus of those who control scientific, technological, and cultural industries to maintain their disproportionate superiority and stifle the voices of others. Adorno and Horkheimer describe the triumph of ideology over equity, discussing the ways by which industrial, commercial, and scientific techniques have been used to apportion and cleanly diversify media channels, population demographics, and economic productivity so that 'none may escape' (Adorno and Horkheimer, 1947: 123). Scientific approaches, they argue, that present entities as objects of normalization and calculation represent the extension of a historical desire to objectify and control. Technologies represent the natural extension of this trajectory, working within a 'circle of manipulation and retroactive need in which the unity of the system grows stronger' (Adorno and Horkheimer, 1947: 121).

If Horkheimer and Adorno are correct when they argue that technologies work to objectify, standardize, and flexibly control, it follows that the Web speaks to power and hierarchy rather than multiplicity and the margins. Yet within influential Western theory lies an alternative whereby technologies can be reconceptualized as performative, embodied, and conversational. Deleuze and Guattari (1987) give voice to a rhizomatic approach toward thinking about ontology, arguing that instead of remaining fixated on objectified and structural forms, one must look at fragments of knowledge as they are performed and embodied, and consider assemblages and alliances that are rootless and unpivoted. The rhizome is flat rather than hierarchical and multiple relations and perspectives are accommodated rather than stifled by an imposed ordering. Marcus Doel (2005: 2) invokes Deleuze and Guattari's insights in arguing that,

> We are not besotted with texts, writings, signs, images and such like . . . reality itself must be lost to us in language: that all we have are signs of things, rather than the things themselves; that having been emancipated from their bondage to an elite band of actually-existing real-world references, signs will at last be free to float in the void, enjoying untroubled and halcyon days.

Applying these insights to the world of new media, I urge us to consider the 'question concerning ontology'. The ontology question considers poetics, aesthetics, and cognition in the creation of devices, databases, and infrastructures. Systems that classify knowledge using databases, that present that knowledge using interfaces, and that network and retrieve that knowledge using algorithms all present opportunities to interrogate the question concerning ontology.

Yet instead of considering diverse cultural practices as a catalyst for re-thinking techne, mainstream pundits instead focus on the importance of increasing global access to pre-created technologies. Thomas Friedman (2005), for example, notes that with the use of networked technologies for outsourcing and the growing interdependence of global corporations, possibilities emerge for greater economic equity.

Yet by focusing on ontology, one can productively complicate Friedman's position and raise several critical questions, including: Is increased access to technology sufficient for creating greater agency, capacity, and empowerment of marginalized voices and agendas? And is access itself as simple as the provision of a technology, or is it instead complicated and potentially compromised by a number of contextual factors?

A focus on the context of use illuminates power dynamics, the question of how uses of a technology may empower agendas held by diverse users. For example, the mobile phone has been seen as the 'killer app' technology for broadening information access and sharing, connecting users and creating horizontal, equal networks. With 4.2 billion users in the year 2009, mobile phones have been seen as a means of improving market efficiency, particularly in agricultural markets (Aker and Mbiti, 2010), fostering access to health information, money (Porteous, 2006), and jobs. Yet further inquiry shows that this narrative is not so straightforward, and that instead mobile networks tend to reify power dynamics based on factors such as who has better access to infrastructure, literacy, and existing social networks (Castells, 1996). Lovink (2010), for example, has pointed out that the lack of finances needed to initiate calls places poorer users 'on call' rather than in a position of autonomy and control. Device users may be unaware of how data gathered about them may be used, and have little ability to control this process. Moreover, due to existing inequalities around infrastructure, literacy, and economic capacity, those already better off may be placed at a position of even higher advantage due to the new technology (Toyama, 2011). The mobile phone is also placed into already existing infrastructures, which themselves are based on the question, 'What values and ethical principles do we inscribe in the inner depths of the built information environment?' (Star, 1999: 379).

Ethnographic research that studies the mobile phone in the context of local environments or infrastructures reveals that technologies tend to be translated, adopted, and shaped as they move locally. For example, Internet ethnographies in Trinidad (Miller and Slater, 2000) and Jamaica (Horst and Miller, 2005) reveal the power held by local value systems and social networks in shaping the destiny of the phone. These studies consider technologies as actors within existing networks of interaction and consider how classic anthropological concepts such as 'kinship' are unhinged to move beyond pre-existing notions of time and space (Caron and Caronia, 2007). These dynamics come about as local communities imprint, shape, and domesticate new technologies.

These insights indicate that the mobile phone, as with many networked technologies, does not simply flatten users into passive consumers. Nor does the fact that they are shaped and acted upon locally automatically turn developing world users into global equals. Instead, by turning toward the study of diverse forms of use, researchers and professionals alike must pay attention to context and ontology, considering important factors such as information environments (Lechner, 1985), which are 'social settings or milieu in which resources, relations, and technologies undergo a structuration-type process of change called informing' (Lievrouw, 2001: 12). These settings include the study of different forms of access, business models, infrastructures (Bowker and Star, 1999), mediating institutions (Heeks, 2008), and already existing capacities which influence the ability for the user or community to act in ways it desires (Appadurai, 2004; Sen, 1995; Srinivasan, 2006a).

Appropriation, authorship, and mobilization

A first step toward thinking about ontology then introduces context, as discussed above, yet even this discussion is limited by the growing reality that many users are not just passive receivers of information, but instead actively involved in social media practices of blogging, commenting, tagging, and remixing (Lessig, 2006). As per Figure 1, how can these acts of appropriation and authorship introduce a further step toward considering the importance of local voices and ontologies in global new media studies?

By carefully analyzing the nature of appropriation, or the process of engaging with a technology, text, or concept and making it one's own, one can identify uses of technology that are generated by communities of users to benefit their own values and priorities, consistent with what Faye Ginsburg (2008: 139) has called 'strategic traditionalism' rather than limiting a discussion of use around the context of access. Silverstone and colleagues (Silverstone et al., 1992: 21), for example, explain that once an object is purchased, or acquired, it enters into a system of 'equivalence and exchange' that re-frames it relative to relational, interpretive, and thresholds of our experience. One's ability to analyze and deconstruct the assumptions and practices associated with a media object must be linked to the awareness of the 'everyday' (Silverstone et al., 1992). While appropriation may not truly empower the sharing of diverse ontologies to the extent where one re-considers technological design and deployment, it still takes the discussion much further than access-oriented research, because it focuses on not just local uses or context, but also the ways in which users creatively subvert and localize technology.

Foundational research in media appropriation has focused on television and film audiences. Despite the argument that television is a passive medium that diminishes a community's 'social capital' (Putnam, 2000), appropriation studies researchers argue that the audience is actively and constructively engaged with the text (which can be written, audiovisual, or interactive). This occurs first through processes of domestication, where technologies are integrated into daily life and environments may change as a result. As Jenkins has illustrated, more active audience members work to organize and construct 'fan cultures', for example creating alternative Star Trek episodes (Jenkins, 1992). The fan is no longer a passive reader but actively constructs an alternative text, creatively engaging with the media artifact (Fiske and Hartley, 1978; Poster, 2008). Baym (2000) has argued that audiences may generate new modes of sociability through acts of re-mediation, blurring online and offline distinctions, and stories that may only loosely cohere with the original. Consistent with this, Ito (2010) presents a tale of Japanese youth and anime that involves active re-use, imagination, and identity construction. These examples present evidence that some distant users could also be seen as 'prosumers' (Jenkins et al., 2005), or consumers who creatively produce and share.

Yet others dismiss these creative acts as ultimately exploitative, free labor that fuels globalized, neoliberal capitalism (Galloway, 2004; Hand and Sandywell, 2002; Terranova, 2000). They argue that the corporation that created the technology gains market value thanks to increased creative use at the margins, and that ultimately the paradigm of corporate control is left unchallenged by the prosumer model. Galloway (2004), for example, argues that Google's value increases thanks to the work of web users, who do the linking, browsing, and page authoring labor needed for its algorithms to exploit. Though there may be more involvement by the user with some new media technologies versus others, the technology's design ultimately circumscribes the nature of its use, and precludes alternate design ontologies.

Yet in some cases technological appropriation may serve as a critical element within a larger grassroots movement. Recent attention has turned to the uses of social media in recent revolutions in Egypt, Tunisia, and other Middle Eastern states (Morozov, 2011). While scholars debate the extent to which Facebook, Twitter, and mobile phones actually impacted these movements, most agree that the unanticipated uses of these technologies by local activists fuel an imaginatory politics that may impact strategy, ideals, and tactics. Technologies, in these cases, often become tools of authorship rather than part of a process of use-based appropriation, interacting with a number of other factors such as community organizations and older media technologies, to generate and sustain a larger-scale movement. A rich history of research in indigenous and community media has provided examples of community-driven authorship, where foreign technologies are subverted for local aims. Faye Ginsburg refers to similar uses by aboriginal communities of new media as 'strategic traditionalism' (2008), arguing:

> The cultural activists creating these new kinds of cultural forms have turned to them as a means of revivifying relationships to their lands, local languages, traditions, and histories and articulating community concerns. They also see media as a means of … revers(ing) processes through which aspects of their societies have been objectified, commodified, and appropriated; their media productions and writings are efforts to recuperate their histories, land rights, and knowledge bases as their own cultural property. (Ginsburg, 2008: 139)

Ginsburg's insights situate technology-facilitated authorship within historical and place-based struggles and aspirations. Her examples, like those of the Arab Spring, are not just about technology, but also entrepreneurial, organizational, social, or political initiatives that incorporate technology authorship into their larger missions. Yet the above examples fail to challenge the actual languages and architectures, or ontologies, of technology. It is this question to which the next section turns.

Ontology, inscription, and shifting the gaze to the network

'Leaky' black boxes

While each of the previous sections discuss important steps toward empowering diverse communities to contextualize, appropriate, and create via new media technologies, none directly consider the machineries of technology, its databases, and networks. While weblogs and video cameras may enable users to create and share stories and perspectives, they are both 'stabilized technologies' (Pinch, 1992), leaving generally invisible the codes, databases, or algorithms by which they are created (Pinch and Bijker, 1984, 1986). This section introduces several approaches toward thinking about new media as mutable, where diverse ontologies form the bases of new media design projects worldwide. By presenting a range of approaches toward thinking about ontology below, new media technologies can be reconceived around the goal of empowering rather than stifling diverse voices.

The above diagram expresses how metaphysics and systems are linked by the ways in which ontology travels from its expression through language into technical systems of storage and retrieval. Ontology is thus not seen as a discipline that exists separately from other forms of inquiry, but instead as the structure of knowledge as induced 'from knowledge embodied in other disciplines' (Corazzon, 2010). Ontologies may speak to propositional and scientific statements about the way the world behaves, cultural and community 'fluid' articulations (Srinivasan and Huang, 2005), modes of bridging different understandings of similar entities, or phenomenological processes and experiences (Polyak and Tate, 1998; Simons 1987; Whitehead, 1929).

Each link in Figure 2 represents a cultural act of interpreting and formalizing knowledge. Ontologies are usually technically instantiated (as in the above diagram) via databases, an artifact of storage that associates that which it carries with the categories by which it is represented. Relational and hierarchical databases, the dominant model of information storage in the computer era, often follow pre-specified protocols toward data storage, persistence, and classification.

The de facto logic behind most ontologies within web systems is of a hierarchical database (see Figure 3). This approach toward ontology presents information according to a set of containers that have 'parent–child' relationships, often carrying rules of attribute inheritance. They express relations between categories through semantically inflexible memberships. Models of classification commonly include hierarchies (most commonly), trees, paradigms, and the flexible though less applied facets (Kwasnik, 1999).

In practice, while this approach toward storing and freezing knowledge in databases allows it to be transferred, replicated, and predictably retrieved, through its seeming neutrality it freezes the ontology around which it is built, thereby reifying cultural, social, and discursive hierarchies. Hope Olson's (2007) analysis of Library of Congress Subject Headings (LCSH), for example, notes the absence of discussion of women and non-Christian religions in the categories presented for classification, pointing to historically perpetuated biases presented in systems seemingly as innocuous as the library. Similar

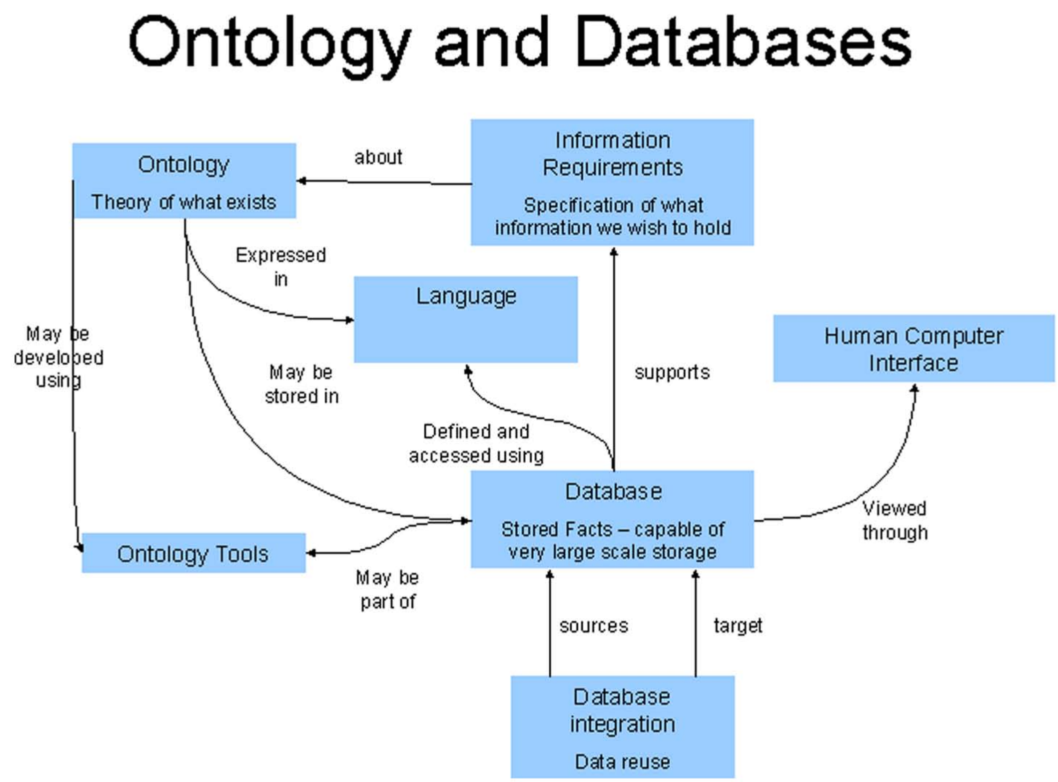

Figure 2. Showing the linkages between philosophies of the world, languages, databases, and technological systems.
Source: ontolog.cim3.net.

findings were uncovered with a group of Native Americans using a fluid ontology model (Srinivasan and Huang, 2005), where community members were presented with the opportunity to semantically model a set of multimedia objects that they had created (Srinivasan, 2007; Srinivasan and Huang, 2005). The topics named and the ways they were grouped in the created 'tree' dramatically differed from how a hierarchical western-created ontology would classify the same set of topics (Srinivasan, 2006b, 2007).

As scholars of classification point out, the ability to find information endorses its right to exist, pointing to the inherent politicization within systems of classification (Bowker and Star, 1999). For well over 150 years, scholars including Melvil Dewey (Wiegand, 1998) and Paul Otlet (Heuvel, 2008) have grappled with the question of how standardized ontologies impact societies, and the means by which users experience 'knowledge'. Olson (2007) points out that these often perpetuate an Aristotelian project by treating categories as discrete, and boundaries of classes as 'watertight'. She notes that these logics must be challenged from poststructural and feminist perspectives, and that we must realize that 'the philosophical tradition of the west has delineated a concept of reason which is exclusive of women and other oppressed groups' (Plumwood, 1993: 436).

Figure 4 is taken from the CIDOC-CRM, an object-oriented domain ontology designed to facilitate the 'interchange of rich and heterogeneous cultural heritage information from museums, libraries and archives' (Gill, 2004). The ontology in Figure 4

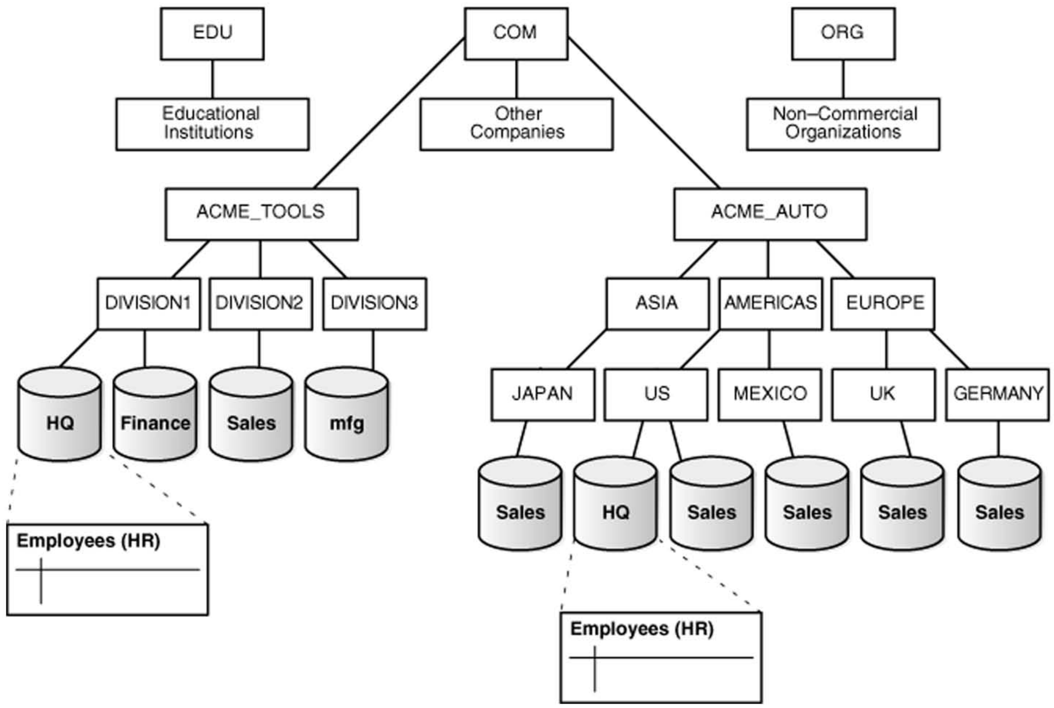

Figure 3. A hierarchical database.
Source: http://download.oracle.com/docs/cd/B28359_01/server.111/b28310/ds_concepts002.htm

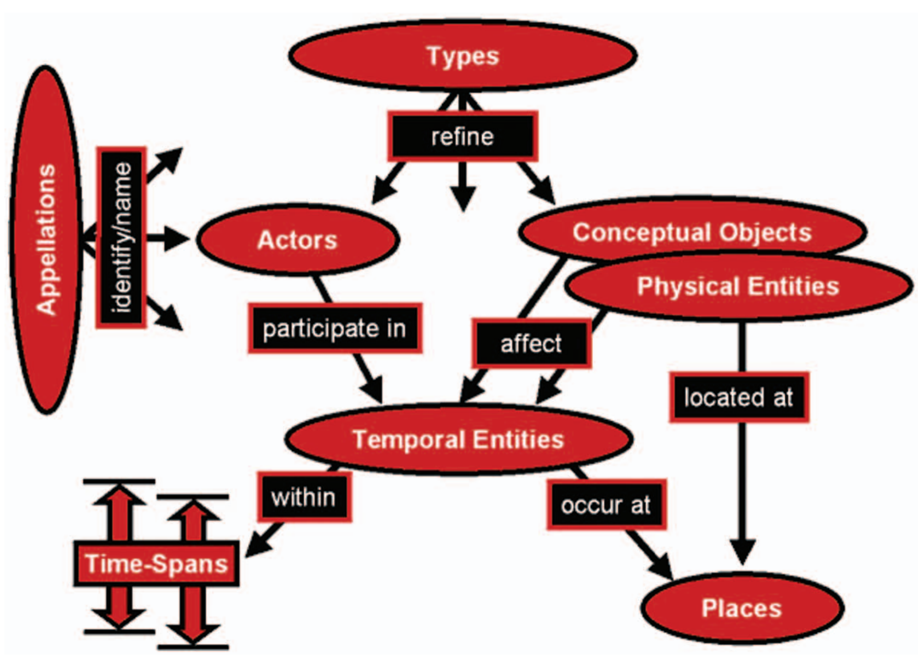

Figure 4. The CIDOC CRM model.
Source: Gill, 2004.

represents an epistemological position applied to a wide variety of cultural objects, originating from communities across the world. Within it, places are not actors, and time and space are disconnected. There is an instantiation of a separation between 'knower and known' in this model, where the cultural object is described by where it stands within a pre-created ontology rather than relative to its cultural context.

Turnbull (1993) points out that the discrete demarcations and divisions that underpin the CIDOC CRM diagram are absent in indigenous ontologies. The ontology illustrated in Figure 4 privileges hierarchy, interoperability, mobility, and scientific abstraction and immutability (Boast et al., 2007), muting other non-Western ways of speaking, describing, and classifying. An indigenous, local knowledge-based ontology in contrast describes, performs, names, and classifies based on ontological concepts excluded from the CIDOC CRM model. While both Western and indigenous ontologies speak to narrating the world, they do so in very different manners that cannot be reconciled into a single ontology (Verran, 2002).

Studying the differences between Cartesian and aboriginal maps and seasonal calendars, Turnbull (1993) points out that discrete demarcations and divisions that underpin the CIDOC CRM diagram are absent in indigenous representations. No time or season ends in totality, nor could either be described as following a simple mathematical probability density-type function. Multiple states exist at once, and all states are present as continuous variables at all times and places. Therefore these ontologies negate a binary or mathematically simple description, instead approaching metaphors that can more easily be understood as improvisational, performative, and complex. To illustrate, Turnbull offers an Australian aboriginal map as an example.

Figure 5 is not obviously recognizable as a map from a classical Cartesian perspective. Yet it is seen and practiced as such by the aboriginal people, functioning as an ontology that allows the Yolngu community to navigate their land, according to their djalkiri, literally translated as 'footprints of the ancestors'. What provides the connection between places are the paths of the ancestral beings. These links determine the conceptions of aboriginal space. And the fact that this space is depicted through graphical notation means that it can be considered a map. The aboriginal map is therefore a narrative depiction that incorporates spatial geography with temporality, order, mythology, and ancestral beings. One does not occupy a binary point of longitudinal and latitudinal intersection wherein the point of occupation is a '1' and all other points are '0'. Instead, occupation is understood as within a set of intersecting songlines, all of which are equal to the others and have been recited by the ancestors.

This example presents an understanding of how diverse cultures and communities maintain divergent knowledge systems, and are thus incommensurable. Lakoff's famous 'Women, Fire, and Dangerous Things' (1987) gives voice to this divergence, when he points out:

> Knowledge, like truth, is relative to understanding. Our folk view of knowledge as being absolute comes from the same source as our folk view that truth is absolute, which is the folk theory that there is only one way to understand a situation. When that folk theory fails, and we have multiple ways of understanding, or 'framing,' a situation, then knowledge, like truth, becomes relative to that understanding. (Lakoff, 1987: 300)

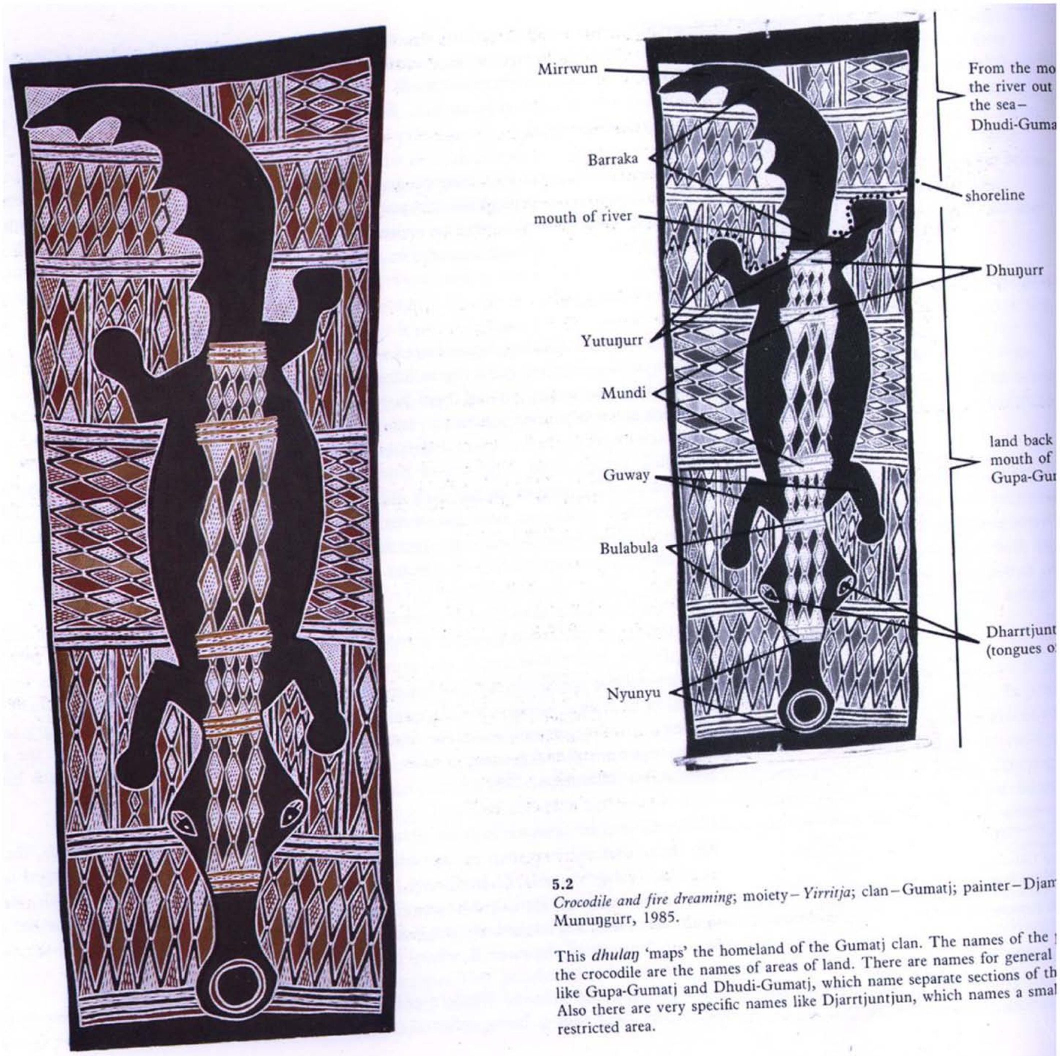

Figure 5. An aboriginal Map ontology.
Source: Turnbull (1993).

What then is the 'contested ontology' of global cyberspace? (Nissenbaum, 2004: 195). Associated with supposed neutral technologies lie a set of economic and social interests, representing the 'background of unexamined cultural assumptions literally designed into the technology itself' (Feenberg, 1995: 87). The standards that drive CIDOC CRM, or hierarchical databases that anchor most web systems, are only controversial while they are in flux, blocking alternative approaches toward thinking how they may have been designed (Bijker and Law, 1992). This insight invokes a classic problem from the history of science, that of how a reliance on computational models creates an experience of automation (Ihde, 1993; Winsberg, 2010), obscuring alternate approaches toward sharing, learning, and knowing. Similarly, Kittler (2009) notes how since Aristotelian times ontologies have formalized rather than explored multiple relationships.

How can global new media projects that work with diverse cultures and communities build on these realizations? In the following three sections, I present a variety of approaches toward working with culturally diverse ontologies.

Fluid ontologies

The fluid ontology approach resists imposing ontologies in lieu of using ethnographic and participatory approaches to collaborate with communities in the design of information systems (Bødker, 1996; Srinivasan and Huang, 2005). A fluid ontology is a set of classifications and semantic relationships expressed by a group of community members used to design a database in a locally appropriate manner. It asks community members to pay explicit attention to their collective, shared practices, or their ethnomethodological activities, and priorities (Srinivasan, 2007). Yet for this approach to truly speak to community, researchers must be mindful that 'community' itself may serve as a homogenized construct, and that often the construction of ontologies reifies elite positions around who is provided with the power to speak within that community (Joseph, 2002; Robbins, 1999).

Ethnomethodology posits that order can be derived from detail, from situated interactions rather than a priori generalizations (Button and Dourish, 1996; Lynch and Peyrot, 1992). Employing these methodological techniques, the fluid ontology approach activates community members to be not just users or Authors, but also designers of systems. The ontology is created through a consensus-driven process where a demographically representative focus group discusses the collective traditions, practices, values, priorities, and epistemologies of their community and through this process generates a semantic map of key topics and their interrelationships. Through this approach (Srinivasan et al., 2005, 2006b, 2007), points of collective importance are identified, and links between these are articulated and dynamically re-cast over multiple community meetings based on new reflections and visions. Scholars have studied the ability to replicate this model through a much larger sample of distributed community users, by sharing randomized pairs of community-identified classifications and asking community members to score the relative semantic similarity between these pairs (Srinivasan, 2009). Figure 6 is an example of a fluid ontology, as created by a set of Native American communities.

Fluid ontologies have been experimented with in several field projects, including the Tribal Peace system, which was designed to connect video, text, and photo contributions between a set of spatially and infrastructurally fragmented Kumeyaay, Luiseno, and Cupeno Native American reservations on the fringes of San Diego County. While this approach directly works with communities to elicit local ontologies and corresponding Authorship and design of databases and classification systems, it neither rejects hierarchy nor the previously criticized tendency to label, conceptualize, and logically formalize. Its main intervention is one of asking local users in environments often ignored to label, organize, and classify.

Process ontologies and the power of experience

While fluid ontologies work with conceptual and entity-based knowledges, mapping and describing relationships between community-articulated themes and topics, they fail to

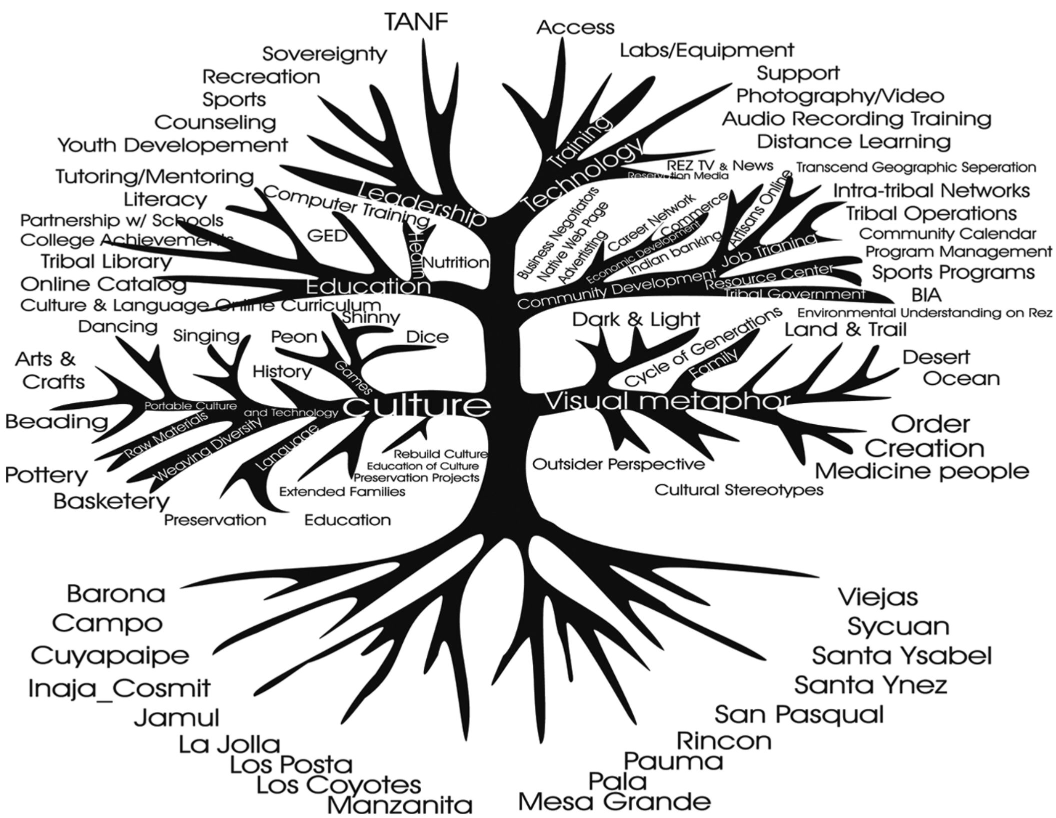

Figure 6. Example of a fluid ontology.
Source: Srinivasan (2007: 1981).

directly engage the process of experience. In contrast, the process ontology model (Polyak and Tate, 1998; Seibt, 1990) links activities with one another, in the context of achieving a theme or task, for example within a business workplace. The common process framework (Polyak and Tate, 1998) has been explicitly developed for businesses, including enterprise, workflow, plan representation, and manufacturing processes.

Outside of this computer-science and business-oriented approach, Whitehead (1929: 77) has discussed the interwoven nature of process and ontology, arguing that our 'actual occasions' of experience structure the universe, and that humans often mistakenly ascribe meaning to entities instead of occasions of experience. Physicist Henry Stapp echoes Whitehead's thinking when he states,

> The world is made of 'actual occasions', each of which arises from potentialities created by prior actual occasions. Actual occasions are 'happenings', each of which comes into being and then perishes, only to be replaced by a successor. These experience-like 'happenings' are the basic realities of nature. (Stapp, 1998: n.p.)

The point made by these writers is that one cannot discuss natural entities without invoking events. Experiences can be analyzed by focusing on perception, which integrates memory and aspiration with the temporal event. These experiences, and the processes that they represent, can be connected and grouped, presenting an ontological approach by which communities can consider the processes by which they act upon and know their worlds.

Multiple ontologies: Embracing networks

Many of the authors reviewed in this article have argued for the truth of incommensurability, stating that no single system or ontology can truly speak to the divergent ontologies held by communities that know and describe the world in diverse ways. The task of working with diverse ontologies is at the core of new research in digital museums (Boast et al., 2007; Srinivasan et al., 2010), and could present a model by which global new media projects can be developed that work with multiple ontologies.

Focusing on building networks rather than databases, the Emergent Databases-Emergent Diversity (ED2) initiative asks indigenous communities, archaeologists, and museum curators to share their insights around objects without any pre-created ontology, and instead embrace one another's perspectives as raw data streams. This experiment has shown how a set of objects excavated from the Zuni Native American reservation of New Mexico is described and classified differently by different 'expert communities', according to diverse incommensurable ontologies. While the Zuni describe the selected objects according to stories and their activities or practices, curators and archaeologists use different categorical and scientific languages that resist being neatly mapped to Zuni insights (Srinivasan et al., 2010). By working with these ontological differences rather than stifling them, the project studies how multiple situated ontologies can exist in parallel and inform one another. Zuni stories, for example, are presented (when permitted by community members) to the community of archaeologists without any translation of what these stories may or may not mean in archaeological language.

As Figure 7 shows, three local systems are networked via an interface that presents 'streams' of shared information 'as is' between the communities, rather than translated or framed a priori. Each system will also respect local protocols held by the expert community regarding the appropriateness of sharing community-held knowledge; making possible, for example, the decision by a Zuni member to only share his or her contributions with the other Zuni members, or perhaps to each of the other expert communities, or to the general public. Local ontologies will also move across different systems as each expert community sees fit, enabling emergent understandings to be formed from multiple, separate ontologies around a given object (Star and Griesemer, 1989).

The approach of working with multiplicity and incommensurable ontologies within ED2 stands in stark contrast to a hierarchical approach that freezes multiple forms of knowledge into a pre-created database logic, and recognizes that biological, linguistic, and cultural diversity emerge through interactions (Turnbull, n.d.) rather than via hierarchical databases. While much of what is accepted as knowledge is in the form of representations: propositions, statements, laws, images, data, equations, graphs, texts, accounts, and so on, the Zuni reveal that knowledge is also embedded in stories, and embodied in practices, activities, rituals, and beliefs, which tend to resist formalization. The ED2 research team has thus argued that to work with these diversities one must

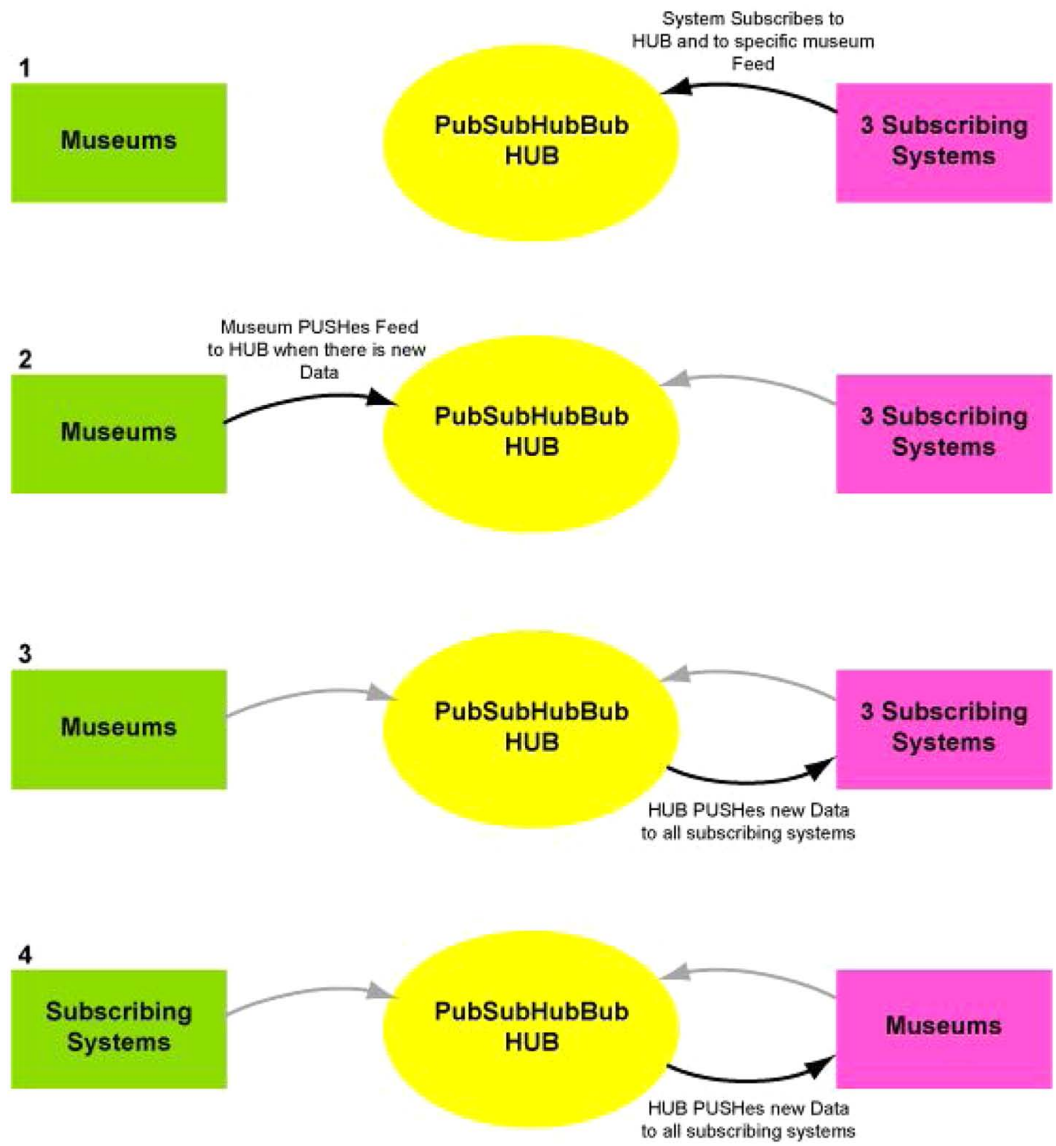

Figure 7. Demonstrating an approach toward working with multiple ontologies through a networked 'hub' that streams information, rather than a continued reliance on databases.

allow each community to speak to one another as is, without any translation, contextualization, or accommodation.

The phenomena of incommensurability need not be solely applied to projects working with multiple cultural groups, but perhaps represents an underlying truth of the web. Weinberger, points out in 'Everything is Miscellaneous: The Power Of The New Digital Disorder' (2008) that with Web 2.0 emerges the opportunity for users to create and share metadata according to their own local experiences and interpretations, thereby stepping away from pre-created ontologies. This represents a philosophical evolution (toward the miscellaneous or local) that steps away from the obsession with a singular, universal semantic mode of describing an information object.

> We've tried to settle on a single, comprehensive framework for knowledge . . . institutions grew to maintain the knowledge framework . . . Because a [new miscellaneous order] is digital, not physical, we no longer have to agree on a single framework. Things have their places, not a single place . . . in the age of miscellaneous (experts) and their institutions are no longer in charge of our ideas. (Weinberger, 2008: 101–102)

Thus, the messiness of the digital world need not mean the end of organization, but just the end of singular, pre-created modes of ordering the world. Rather than stifle diversity, embrace the truth of miscellany.

Conclusions

This article has argued that global new media researchers and designers must continue to embrace culturally diverse ontologies by interrogating the languages, architectures, and infrastructures with which they work. Figure 1 distills the argument that as computers and mobile phones circulate to billions worldwide, passive information access must be discarded in lieu of more culturally situated research projects. It is important that social and political debates introduce the challenge and dilemmas of ontology this article points to, placing them side-by-side with important public issues around net neutrality, increased access, and copyright and fair use issues. If studies of passive use continue to dominate new media studies research, we will continue to ignore environment and context, and at best consider appropriation and authorship without challenging the mutability of technology. A renewed focus on ontology reminds us that networks, algorithms, and databases are social and cultural constructs, and that diverse ways of knowing and speaking can inspire rather than confound global new media initiatives.

Further research by practitioners and scholars can push these cultural codes further. For example, have we reached the point where databases should be discarded in lieu of approaches that trust our ability to search and make sense of disorienting amounts of data? Is the Deleuzian approach of moving away from metaphor, concept, word, and category realizable in such a world? And are websites such as Chatroulette.com, or projects such as ED2, good examples of privileging the performative and experiential? If we embrace miscellany and multiple ontologies, how do we ensure that we can still access and find information without being stifled by potential biases built into filter bubbles (Pariser, 2011a, 2011b) or search algorithms (Introna and Nissenbaum, 2000)? Srinivasan (2009) has, for example, made the point that previous Google searches for 'Africa' returned very few primary results from web-pages based out of that continent, speaking to the PageRank algorithm's bias toward the parts of the world which feature more robust access, infrastructure, wealth, and literacy. Do these ontological biases speak to the continued reign of those with power worldwide? Can 'progressive algorithms' which speak to the multiplicity of value systems and priorities worldwide be developed and equitably deployed?

Embracing incommensurability is a turn away from many years of labeling, classification, and the reliance on databases as fixed, hierarchical ways of modeling knowledge. Embracing multiple and local ontologies can allow media studies researchers and practitioners to embrace the digital world's potential to diverse community and cultural voices.

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